

PartC – Inheritance in Java Lab Report

1. Aim

To develop a Java program demonstrating inheritance by creating a parent class person and two child classes, student and faculty, that inherit the properties and methods of the parent class.

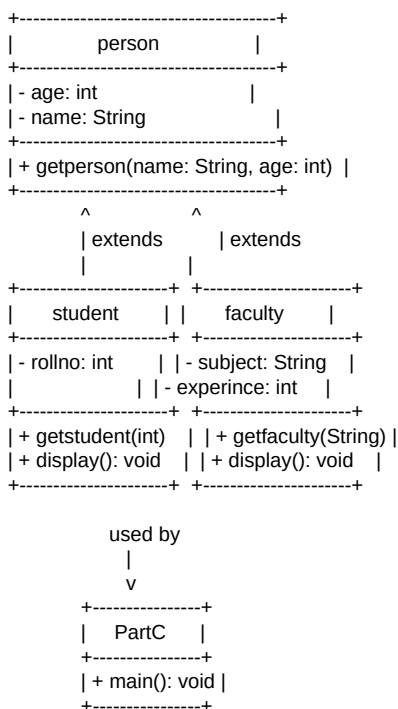
2. Description

The program belongs to the package **tasktwo**. The person class contains the common data members age and name, along with the getperson() method. The student and faculty classes extend person and inherit its fields and method. The student class adds rollno and the getstudent() method, while the faculty class adds subject and the getfaculty() method. Both child classes define their own display() method to print their respective details. The PartC class creates objects of student and faculty and displays their information.

3. Tools Used

- Programming Language: Java
- Development Environment: Any Java IDE or text editor
- Execution Tool: Java Development Kit (JDK)
- Concepts Used: Inheritance, superclass, subclass, method, object creation, this keyword, and method overriding-style behavior

4. UML Class Diagram



5. Algorithm

1. Start the program.
2. Create the person class with age and name fields and define getperson() to assign their values.
3. Create the student class by extending person.
4. Add rollno and getstudent() to the student class and define display() to print student details.

5. Create the faculty class by extending person.
6. Add subject and getfaculty() to the faculty class and define display() to print faculty details.
7. Create a student object and assign the student's name, age, and roll number.
8. Create a faculty object and assign the faculty member's name, age, and subject.
9. Call display() for both objects.
10. Stop the program.

6. Complete Java Source Code

```
package tasktwo;

class person {
    int age;
    String name;

    void getperson(String name, int age) {
        this.age = age;
        this.name = name;
    }
}

class student extends person {
    int rollno;

    void getstudent(int rollno) {
        this.rollno = rollno;
    }

    void display() {
        System.out.println("Age is :" + this.age);
        System.out.println("Name is :" + this.name);
        System.out.println("Rollno is:" + this.rollno);
    }
}

class faculty extends person {
    String subject;
    int experince;

    void getfaculty(String subject) {
        this.subject = subject;
    }

    void display() {
        System.out.println("Subject is :" + this.subject);
        System.out.println("Name is:" + this.name);
        System.out.println("age is:" + this.age);
    }
}

public class PartC {
    public static void main(String[] args) {
        student s = new student();
        s.getperson("Srikar", 19);
        s.getstudent(238);

        faculty f = new faculty();
        f.getperson("santhosh", 43);
        f.getfaculty("java");

        s.display();
        f.display();
    }
}
```

7. Sample Output

```
Age is :19  
Name is :Srikar  
Rollno is:238  
Subject is :java  
Name is:santhosh  
age is:43
```

8. Result

The Java program successfully demonstrates single-level inheritance from the person class to the student and faculty classes. The child classes reuse the parent class's age, name, and getperson() method while adding their own properties and display methods. The program displays the details of both a student and a faculty member.